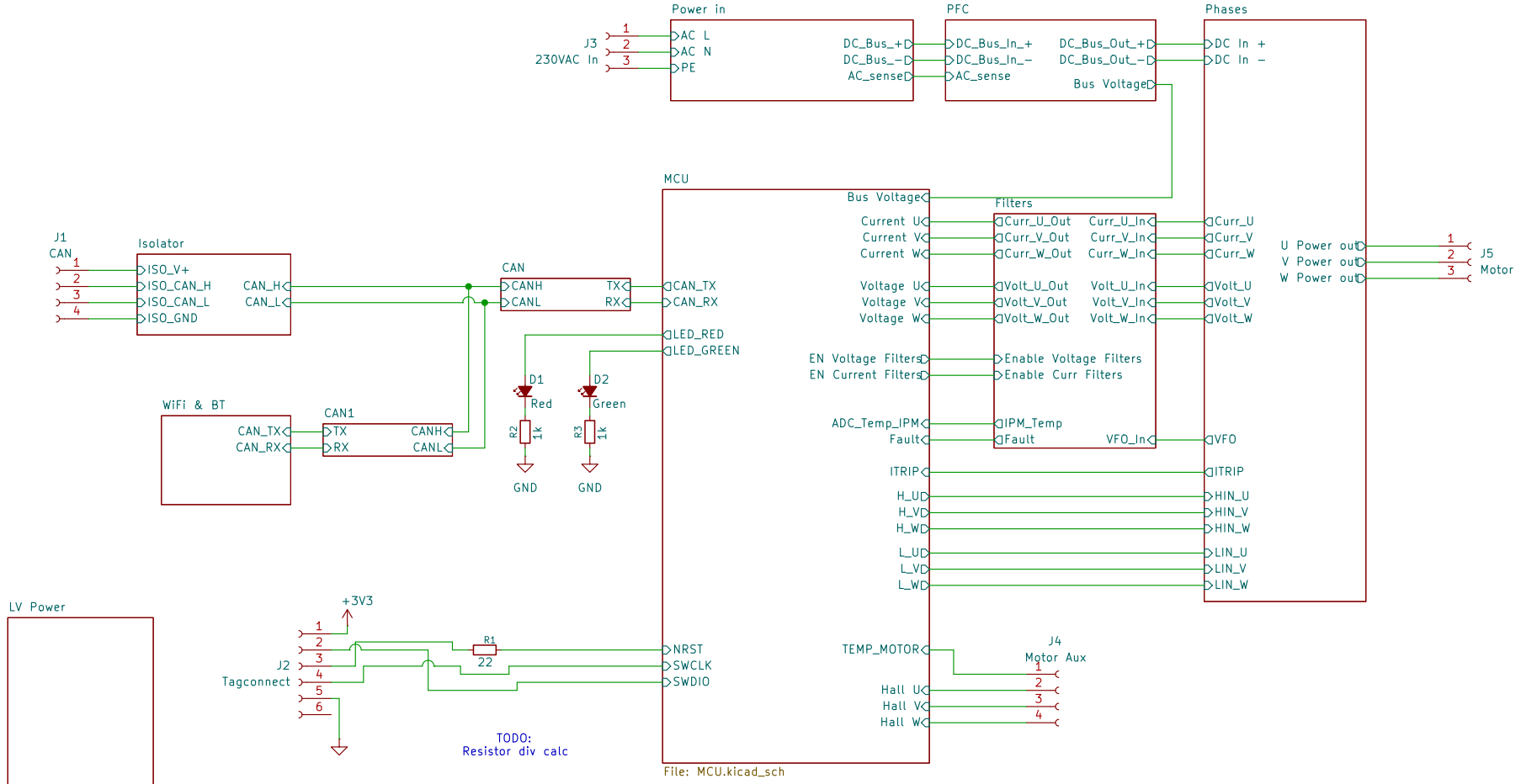


HVAC VESC

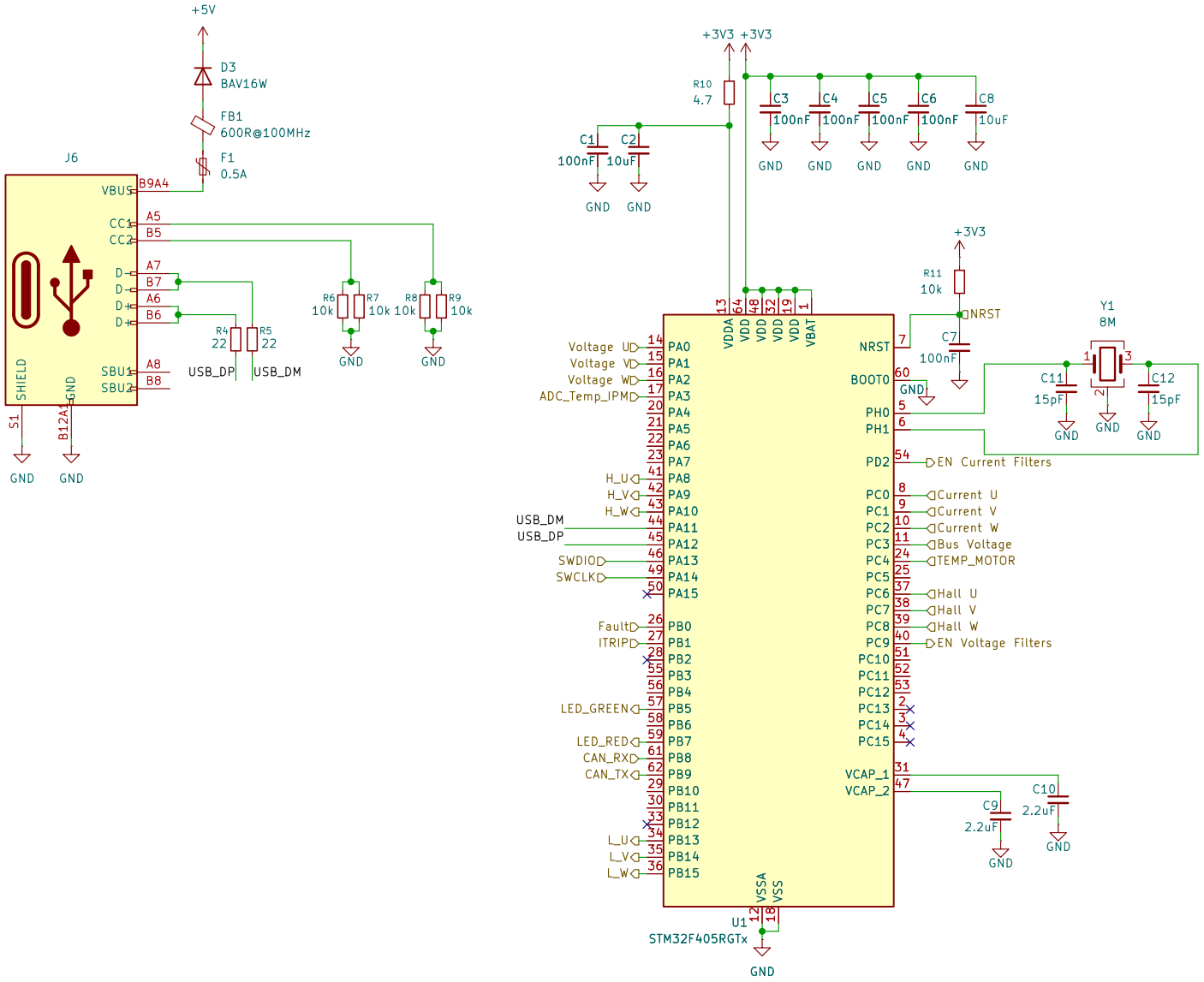
Needs to follow for CE cert:
EN IEC 63000 (RoHS)
IEC 60335-2-34 (Particular requirements for motor-compressors)
IEC_61000-3-2 (Limits for harmonic current emissions) <--- Most important

Picobello B.V.

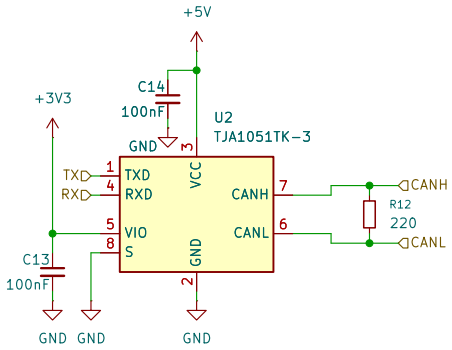
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File: CompressorController.kicad_sch

Title: Mains Powered VESC MD

Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 1/11



Picobello B.V.		
Sheet: /MCU/		
File: MCU.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 2/11



Picobello B.V.		
Sheet: /CAN1/		
File: can.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 3/11

DC In +

DC In −

DC Bus +

DC Bus −

J8
XT90
1
2
3
4
GND

Development plug
for injecting DC voltage

+15V

C18
2.2uF
GND

C19
100nF
GND

C23
1nF
GND

U6
IKCM10H60GAXKMA1

VDD 13
VSS 16
NC 24
GND
VS(U) 1
VB(U) 2
C20 10uF
C24 100nF
VS(V) 3
VB(V) 4
C21 10uF
C25 100nF
VS(W) 5
VB(W) 6
C22 10uF
C26 100nF
HIN(U) 7
HIN(V) 8
HIN(W) 9
LIN(U) 10
LIN(V) 11
LIN(W) 12
VFO 14
ITRIP 15
P 23
U 22
V 21
W 20
NU 19
NV 18
NW 17

HIN_UD R13
HIN_VD R14
HIN_WD R15
LIN_UD R16
LIN_VD R17
LIN_WD R18
R19 10k
R20 10k
R21 10k
R22 10k
R23 10k
R24 10k
GND

VFO
ITRIP
1k R25
C27
1nF
GND

DC Bus +
C28
470nF

IKCM10H60GA
C3615235

+3V3
C15
100nF
GND
TP7
Curr_U
U3
INA240A1D
GND

+3V3
C16
100nF
GND
TP8
Curr_V
U4
INA240A1D
GND

+3V3
C17
100nF
GND
TP9
Curr_W
U5
INA240A1D
GND

R26 50mR
R27 50mR
R28 50mR

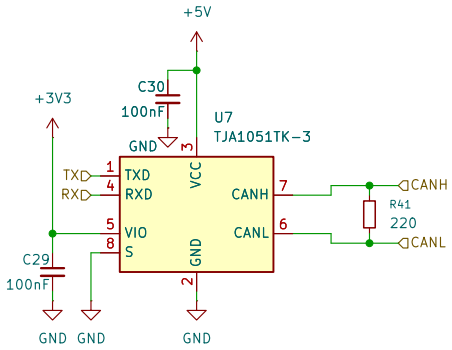
DC Bus −

U Power out
200k R29
200k R30
200k R31
Volt_U
3.16k R32
GND

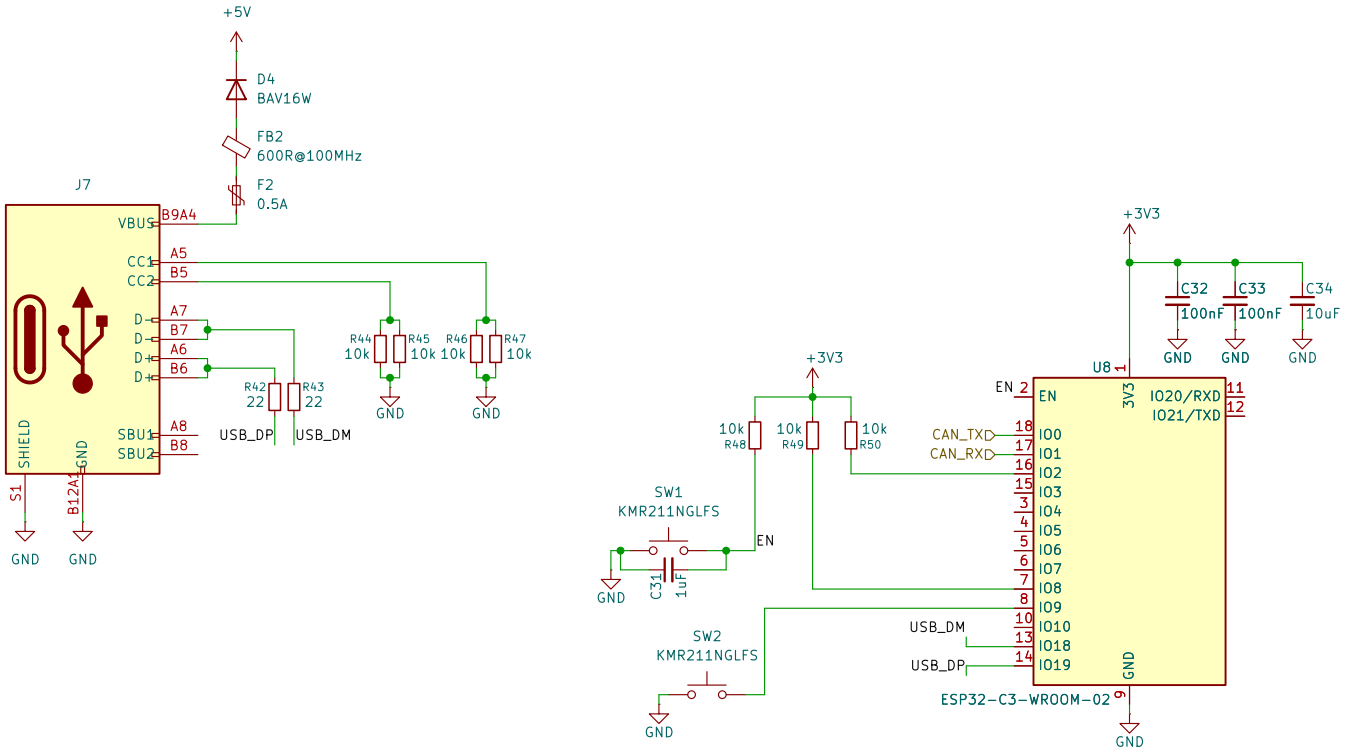
V Power out
200k R33
200k R34
200k R35
Volt_V
3.16k R36
GND

W Power out
200k R37
200k R38
200k R39
Volt_W
3.16k R40
GND

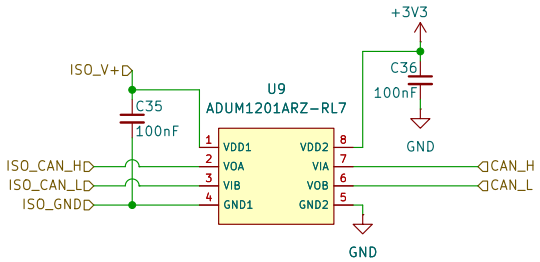
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Sheet: /Phases/
File: phases.kicad_sch
Title: Mains Powered VESC MD
Size: A3 | Date: 2026-04-16 | Rev: R1
KiCad E.D.A. 10.0.0 | Id: 4/11



Picobello B.V.		
Sheet: /CAN/		
File: can.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 5/11

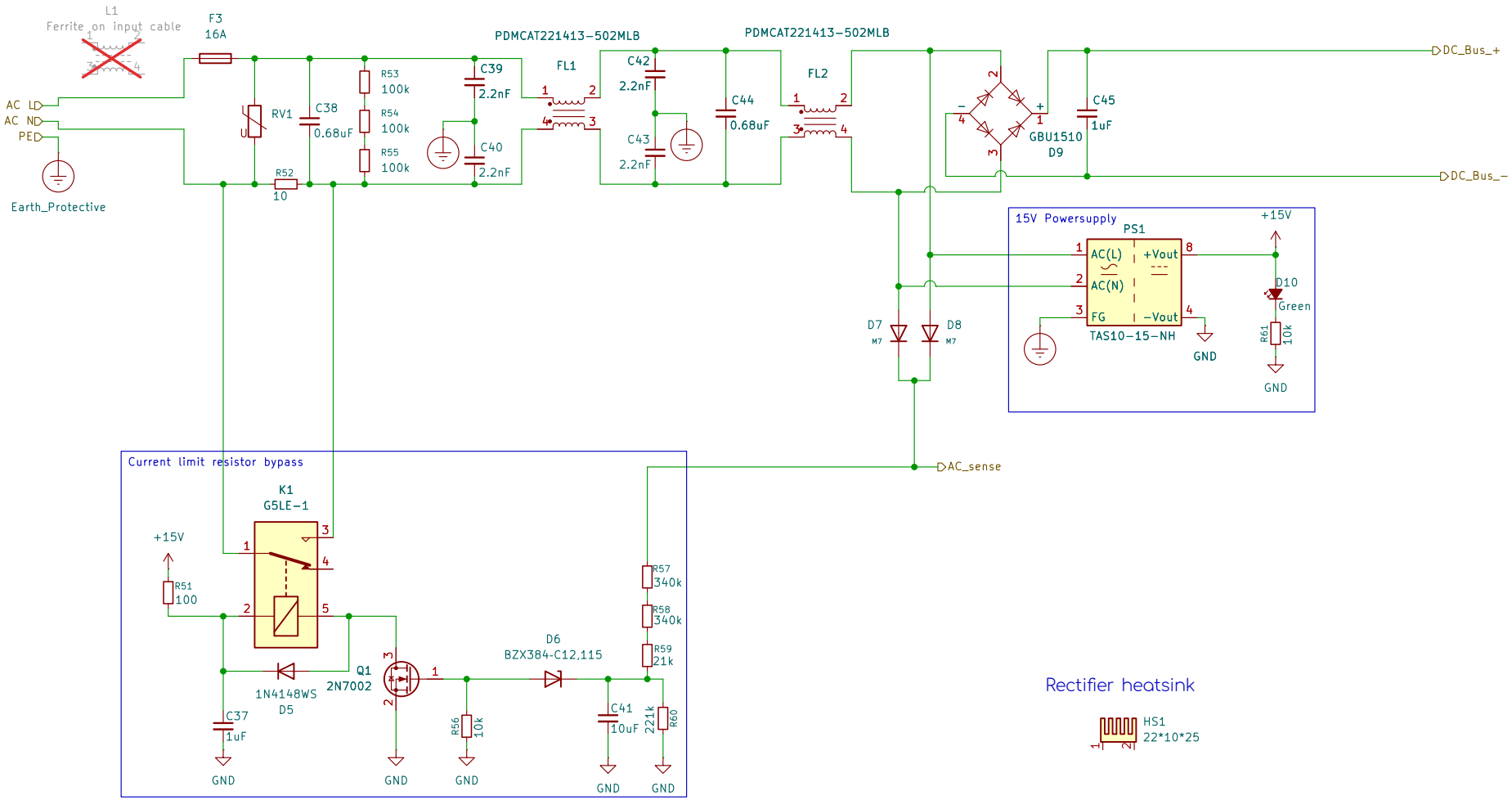


Picobello B.V.		
Sheet: /WiFi & BT/		
File: express.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 6/11



Picobello B.V.		
Sheet: /Isolator/		
File: isolator.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 7/11

Following TIDA-00443



Picobello B.V.		
Sheet: /Power in/		
File: powerin.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 8/11

Following TIDA-00443

Heatsink for D14

D12
1N5406

L2
330uH

D14
DSEP30-06A

HS2
XSD683-022

R77
43.2

C69
330pF

DC_Bus_In_+

DC_Bus_In_-

DC_Bus_Out_+

DC_Bus_Out_-

C51
1.5uF

R69
0.2

R68
220

R70
0.02

IDk why, but TI ref design does this, trim?

Gate drive

U16
UCC27517ADBV

+15V

C55
10uF

C56
100nF

GND

D13
MMSD914

D18
MMSD914

D19
MMSD914

R74
2

R75
1

R76
10k

Q3
IPL60R185C7

R98
2

R99
1

Q4
IPL60R185C7

R101
2

R102
1

Q5
IPL60R185C7

C72
220uF

C73
220uF

C75
220uF

R84
200k

R85
200k

R86
200k

R87
3.16k

Bus Voltage

R92
221k

R94
221k

Bleeder boys

R93
221k

R95
221k

TI's Fet got quite hot (90C)
So 3x parallel, and better FETs as well

PFC controller

U15
UCC28180D

GND
ICOMP
ISENSE
FREQ
GATE
VCC
VSENSE
VCOMP

+15V

C54
2.7nF

R71
21.5k

R73
23.7k

C61
4.7uF

C62
390nF

C64
100nF

C65
10uF

TP5
HVDC-Sense

R88
332k

R89
332k

R90
340k

R91
13k

C76
820pF

Boost follower control and sensing

AC_sense

R62
332k

R63
332k

R64
340k

TP6
ACSense

AC Sense

R65
10k

C46
10uF

D11
BAT54J

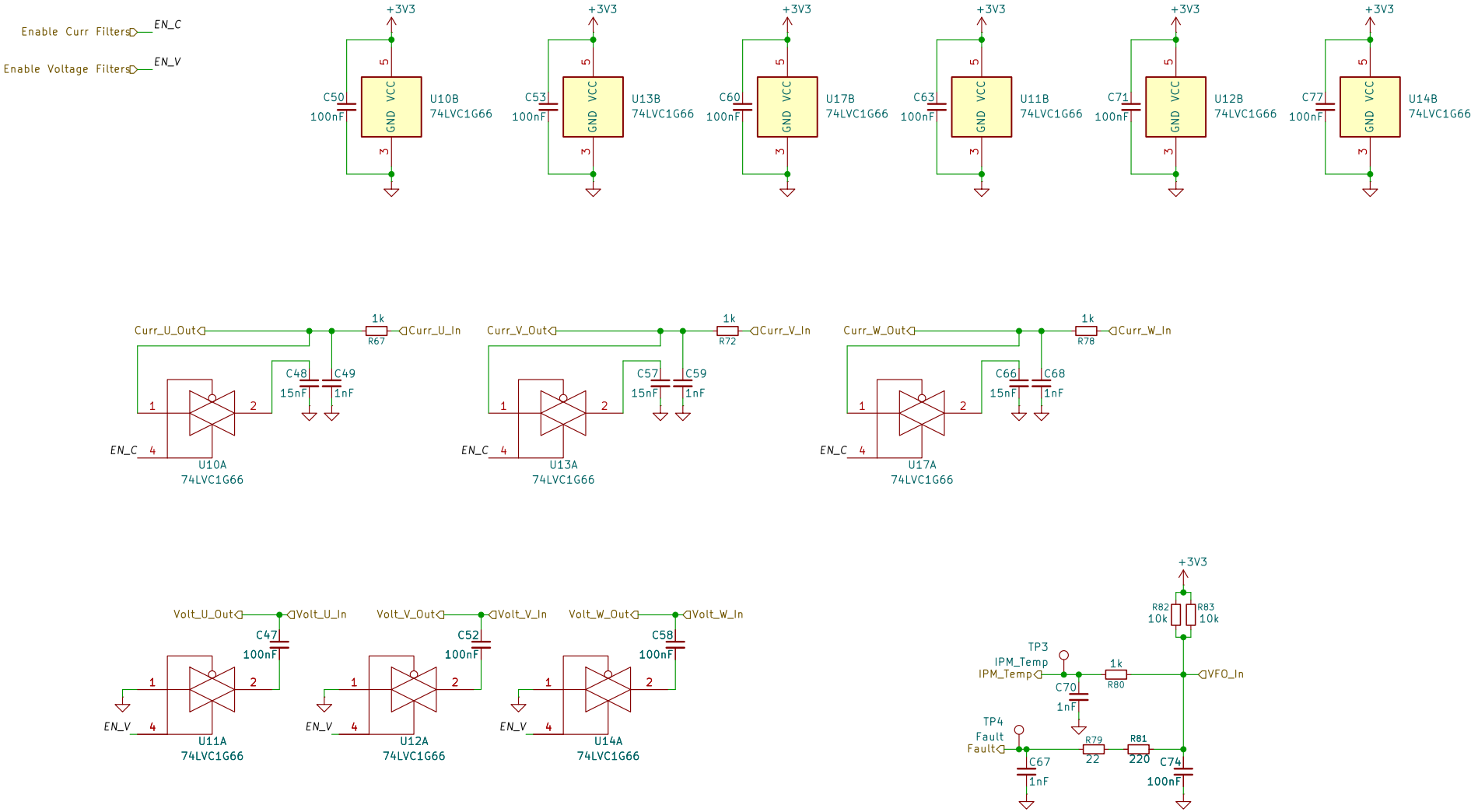
Vsense

Q2
BC847C

R66
13.7k

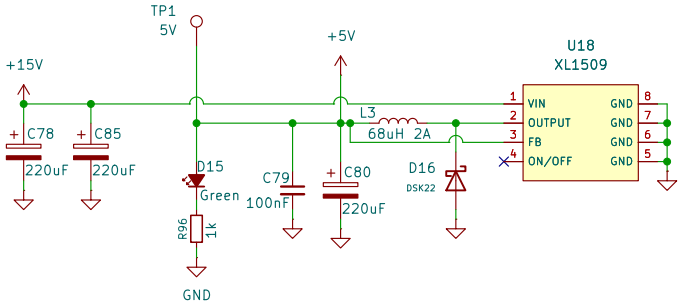
Boost follower is actually quite a bit more efficient.
But the software might not like the regulation loop.

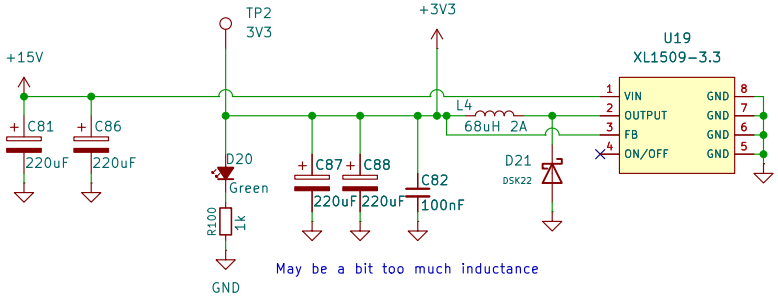
Picobello B.V.	
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File: pfc.kicad_sch	
Title: Mains Powered VESC MD	
Size: A3	Date: 2026-04-16
KiCad E.D.A. 10.0.0	Rev: R1
	Id: 9/11



VFO Pin is also UVLO and OC fault output

Picobello B.V.	
Sheet: /Filters/	
File: filters.kicad_sch	
Title: Mains Powered VESC MD	
Size: A4	Date: 2026-04-16
KiCad E.D.A. 10.0.0	**Rev: R1**
	Id: 9/11





Picobello B.V.		
Sheet: /LV Power/		
File: lvpower.kicad_sch		
Title: Mains Powered VESC MD		
Size: A4	Date: 2026-04-16	**Rev: R1**
KiCad E.D.A. 10.0.0		Id: 10/11